Data Visualization using matplotlib

Problem Statement: Analyzing Air Quality Index (AQI) Trends in a City Dataset: "City_Air_Quality.csv" Description: The dataset contains information about air quality measurements in a specific city over a period of time. It includes attributes such as date, time, pollutant levels (e.g., PM2.5, PM10, CO), and the Air Quality Index (AQI) values. The goal is to use the matplotlib library to create visualizations that effectively represent the AQI trends and patterns for different pollutants in the city. Tasks to Perform:

1. Import the "City_Air_Quality.csv" dataset.
2. Explore the dataset to understand its structure and content.
3. Identify the relevant variables for visualizing AQI trends, such as date, pollutant levels, and AQI values.
4. Create line plots or time series plots to visualize the overall AQI trend over time.
5. Plot individual pollutant levels (e.g., PM2.5, PM10, CO) on separate line plots to visualize their trends over time.
6. Use bar plots or stacked bar plots to compare the AQI values across different dates or time periods.
7. Create box plots or violin plots to analyze the distribution of AQI values for different pollutant categories.
8. Use scatter plots or bubble charts to explore the relationship between AQI values and pollutant levels.

Code :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

1. Import the "City_Air_Quality.csv" dataset.

```
df=pd.read_csv("city_day.csv")
df.head()
```

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucke
0	Ahmedabad	2015-01-01	NaN	NaN	0.92	18.22	17.15	NaN	0.92	27.64	133.36	0.00	0.02	0.00	NaN	Nat
1	Ahmedabad	2015-01-02	NaN	NaN	0.97	15.69	16.46	NaN	0.97	24.55	34.06	3.68	5.50	3.77	NaN	Nat
2	Ahmedabad	2015-01-03	NaN	NaN	17.40	19.30	29.70	NaN	17.40	29.07	30.70	6.80	16.40	2.25	NaN	Nat
3	Ahmedabad	2015-01-04	NaN	NaN	1.70	18.48	17.97	NaN	1.70	18.59	36.08	4.43	10.14	1.00	NaN	Nat
4	Ahmedabad	2015-01-05	NaN	NaN	22.10	21.42	37.76	NaN	22.10	39.33	39.31	7.01	18.89	2.78	NaN	Nat

2. Explore the dataset to understand its structure and content.

df.columns

```
Index(['City', 'Date', 'PM2.5', 'PM10', 'NO', 'NO2', 'NOx', 'NH3', 'CO', 'SO2',
       'O3', 'Benzene', 'Toluene', 'Xylene', 'AQI', 'AQI_Bucket'],
      dtype='object')
```

df.shape

(29531, 16)

df.isnull().sum()

```
City              0
Date              0
PM2.5          4598
PM10          11140
NO             3582
NO2            3585
NOx            4185
NH3           10328
CO             2059
SO2            3854
O3             4022
Benzene        5623
Toluene        8041
Xylene        18109
AQI            4681
AQI_Bucket     4681
dtype: int64
```

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   City        29531 non-null  object
 1   Date        29531 non-null  object
 2   PM2.5       24933 non-null  float64
 3   PM10        18391 non-null  float64
 4   NO          25949 non-null  float64
 5   NO2         25946 non-null  float64
 6   NOx         25346 non-null  float64
 7   NH3         19203 non-null  float64
 8   CO          27472 non-null  float64
 9   SO2         25677 non-null  float64
 10  O3          25509 non-null  float64
 11  Benzene     23908 non-null  float64
 12  Toluene     21490 non-null  float64
 13  Xylene      11422 non-null  float64
 14  AQI         24850 non-null  float64
 15  AQI_Bucket  24850 non-null  object
dtypes: float64(13), object(3)
memory usage: 3.6+ MB
```

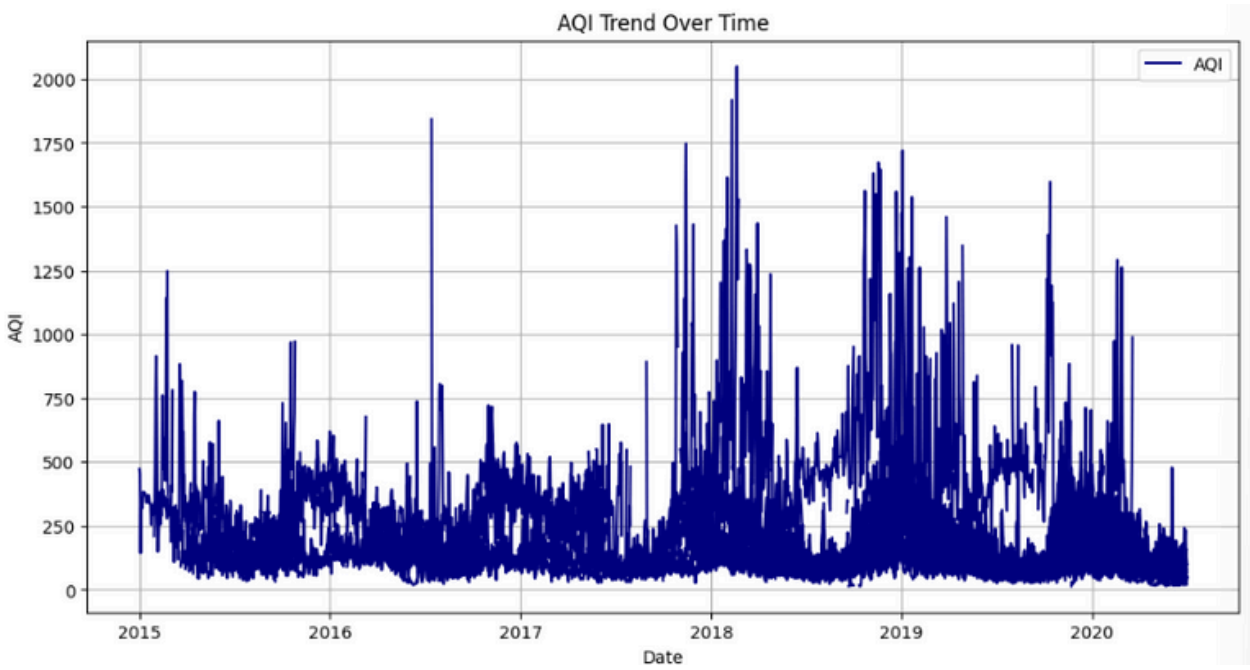
```
# Convert Date to datetime type for plotting
df['Date'] = pd.to_datetime(df['Date'])
```

3. Identify the relevant variables for visualizing AQI trends, such as date, pollutant levels, and AQI values.

- **Date**
- **Pollutants: PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene, Xylene**
- **AQI**
- **AQI_Bucket**

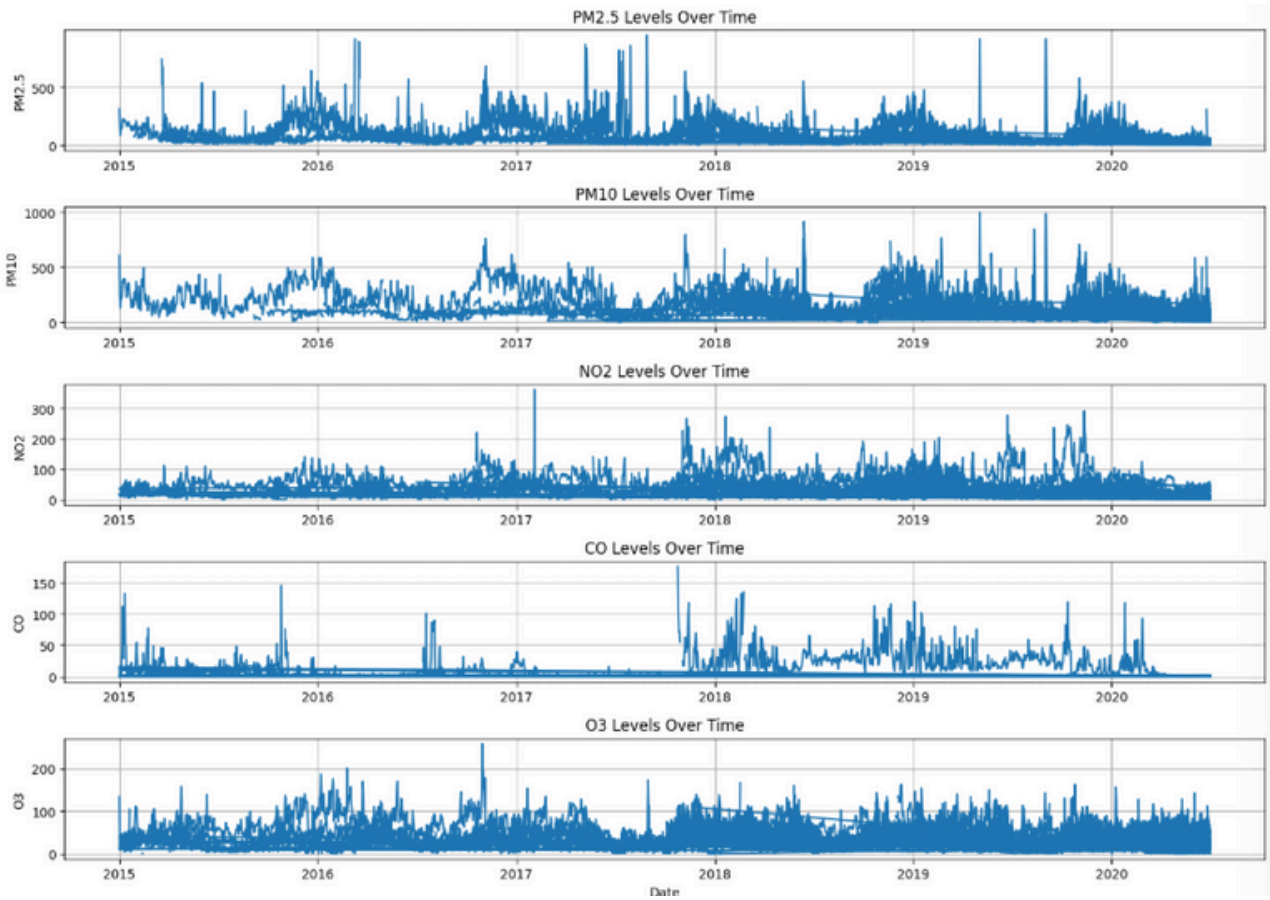
4. Create line plots or time series plots to visualize the overall AQI trend over time.

```
plt.figure(figsize=(12,6))
plt.plot(df['Date'], df['AQI'], color='navy', label='AQI')
plt.xlabel('Date')
plt.ylabel('AQI')
plt.title('AQI Trend Over Time')
plt.legend()
plt.grid(True)
plt.show()
```



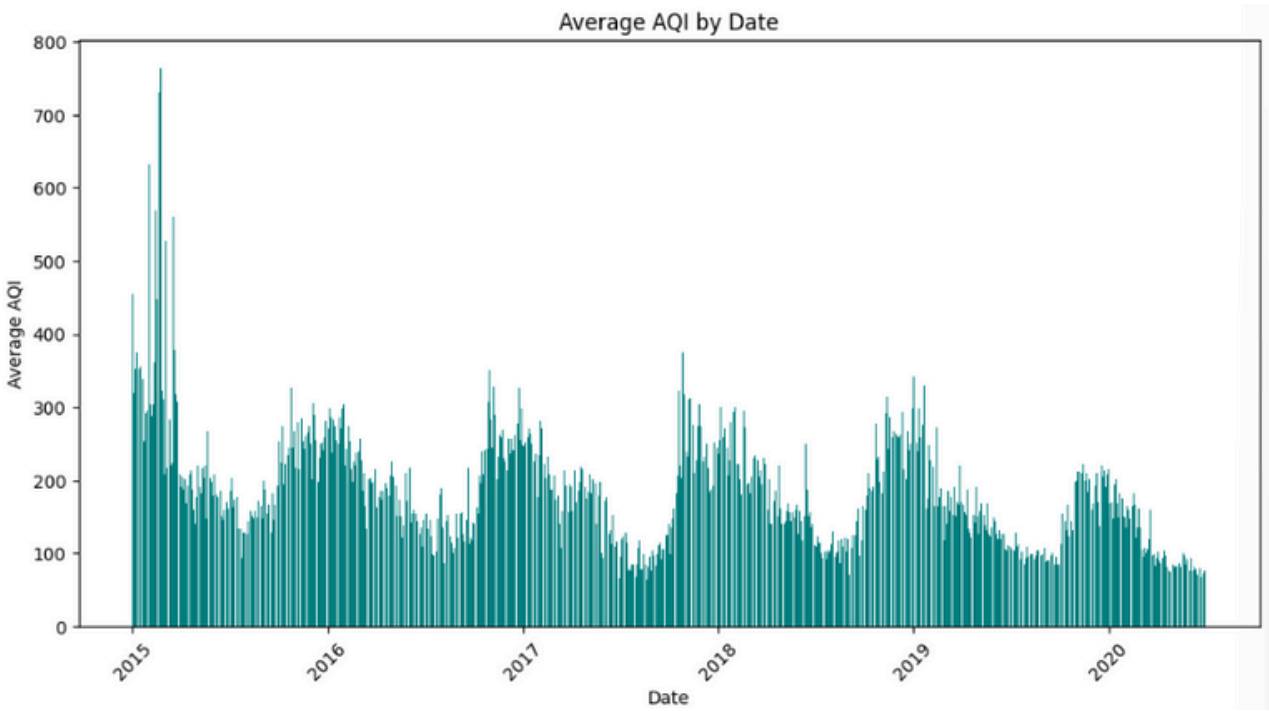
5. Plot individual pollutant levels (e.g., PM2.5, PM10, CO) on separate line plots to visualize their trends over time.

```
plt.figure(figsize=(14,10))

pollutants = ['PM2.5', 'PM10', 'NO2', 'CO', 'O3']

for i, pollutant in enumerate(pollutants, 1):
    plt.subplot(len(pollutants), 1, i)
    plt.plot(df['Date'], df[pollutant], label=pollutant)
    plt.ylabel(pollutant)
    plt.title(f'{pollutant} Levels Over Time')
    plt.grid(True)

plt.xlabel('Date')
plt.tight_layout()
plt.show()
```



6. Use bar plots or stacked bar plots to compare the AQI values across different dates or time periods

```
daily_avg_aqi = df.groupby('Date')['AQI'].mean().reset_index()

plt.figure(figsize=(12,6))
plt.bar(daily_avg_aqi['Date'], daily_avg_aqi['AQI'], color='teal')
plt.xlabel('Date')
plt.ylabel('Average AQI')
plt.title('Average AQI by Date')
plt.xticks(rotation=45)
plt.show()
```



7. Create box plots or violin plots to analyze the distribution of AQI values for different pollutant categories.

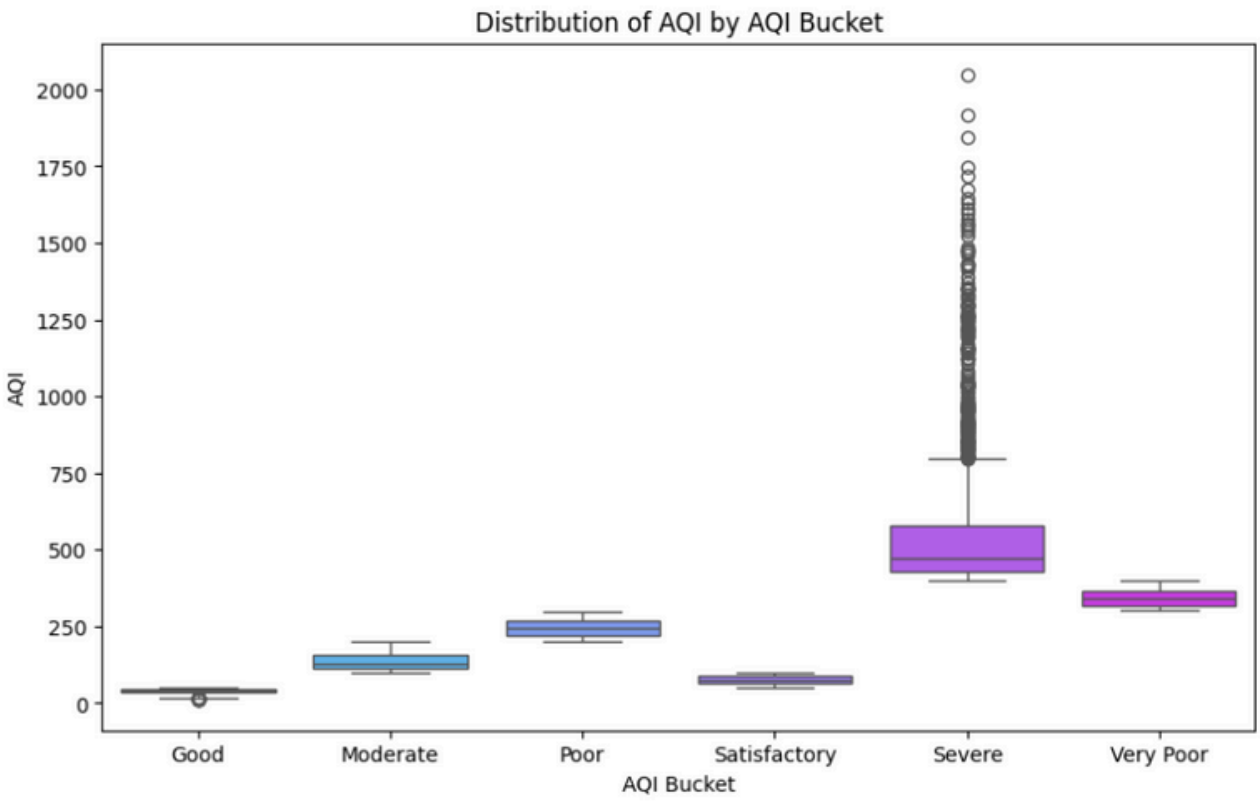
print(df['AQI_Bucket'].unique())

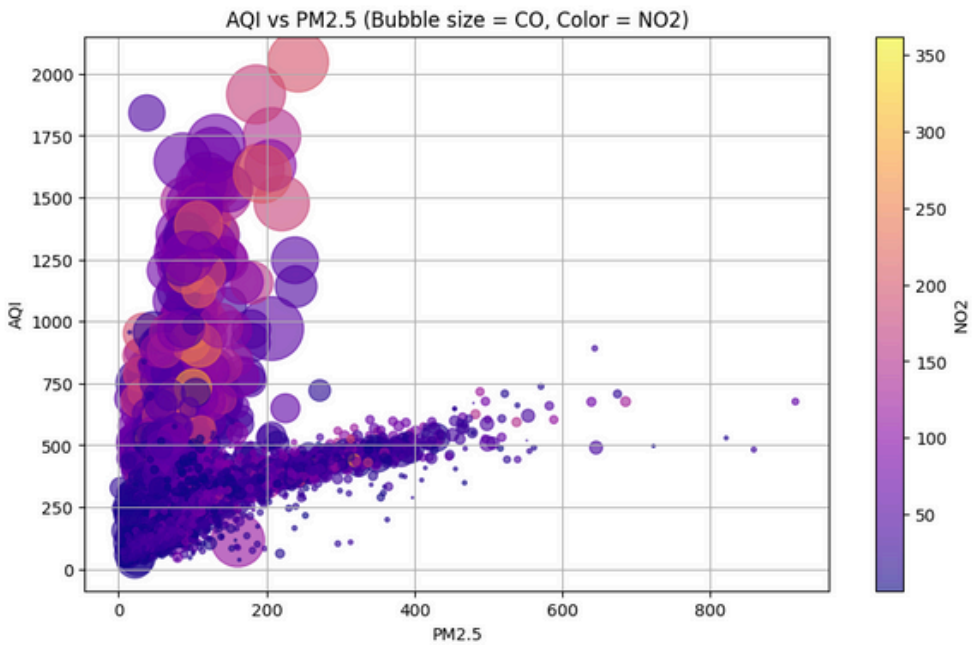
```
[nan 'Poor' 'Very Poor' 'Severe' 'Moderate' 'Satisfactory' 'Good']
```

print(df['AQI_Bucket'].apply(type).value_counts())

```
AQI_Bucket
<class 'str'>      24850
<class 'float'>     4681
Name: count, dtype: int64
```

```
# Convert to string (this converts NaNs to 'nan' string)
df['AQI_Bucket'] = df['AQI_Bucket'].astype(str)

# Filter out rows where 'AQI_Bucket' is 'nan'
df_filtered = df[df['AQI_Bucket'] != 'nan']

# Sort categories alphabetically (or define a custom order if you want)
categories = sorted(df_filtered['AQI_Bucket'].unique())

plt.figure(figsize=(10,6))
sns.boxplot(x='AQI_Bucket', y='AQI', data=df_filtered, palette='cool', order=categories)
plt.title('Distribution of AQI by AQI Bucket')
plt.xlabel('AQI Bucket')
plt.ylabel('AQI')
plt.show()
```



8. Use scatter plots or bubble charts to explore the relationship between AQI values and pollutant levels.

```
plt.figure(figsize=(10,6))
scatter = plt.scatter(df['PM2.5'], df['AQI'], s=df['CO']*10, c=df['NO2'], cmap='plasma', alpha=0.6)
plt.xlabel('PM2.5')
plt.ylabel('AQI')
plt.title('AQI vs PM2.5 (Bubble size = CO, Color = NO2)')
plt.colorbar(scatter, label='NO2')
plt.grid(True)
plt.show()
```